Additional Post-Hoc Noise Robustness Analysis
Overall classification under sensor measurement noise

Noise	Coverage	RF Macro-F1	LightGBM Macro-F1	RF HW macro recall	LightGBM HW macro recall
Clean	100.0000%	0.9505	0.9021	0.9968	0.8305
1%	92.1941 ± 0.0737%	0.9386 ± 0.0003	0.8872 ± 0.0005	0.9949 ± 0.0007	0.8284 ± 0.0007
2%	84.4133 ± 0.0997%	0.9185 ± 0.0008	0.8528 ± 0.0011	0.9762 ± 0.0013	0.8080 ± 0.0019
5%	74.7809 ± 0.1046%	0.8519 ± 0.0009	0.7311 ± 0.0015	0.8934 ± 0.0029	0.6852 ± 0.0027
10%	69.5123 ± 0.1124%	0.7066 ± 0.0014	0.5693 ± 0.0023	0.7525 ± 0.0039	0.5370 ± 0.0036

Coverage is identical for RF and LightGBM; one shared column. Noisy values: mean ± sample SD across 10 paired seeds.

Coverage decreases because the unchanged input-quality rules can convert noise-induced negative electrical readings to missing values and trigger abstention.

End-to-end pipeline robustness under Gaussian sensor noise, including existing input-quality/abstention rules.